ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MATHEMATICAL LITERACY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mathematical Literacy paper 1 · practice a.

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How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods and sensible explanations.

QUESTION 1

1.1 One withdrawal: R8,50 + 15 × R1,20 = R26,50 ✓✓; two withdrawals = R53,00 ✓; debit orders 3 × R4,50 = R13,50 ✓; total = 65 + 53 + 13,50 = R131,50 ✓ (5)

1.2.1 850 × 23,40 ✓ = R19 890 ✓ (2)

1.2.2 Commission = 19 890 × 0,015 = R298,35 ✓; total = R20 188,35 ✓ (2)

1.3 180 000 × $(1{,}055)^3$ ✓✓ = R211 363,45 ✓ (3)

1.4.1 Profit per vetkoek = 8,00 − 3,50 = R4,50 ✓; 1 800 ÷ 4,50 ✓ = 400 vetkoek ✓ (3)

1.4.2 Income = 650 × 8 = R5 200; cost = 1 800 + 650 × 3,50 = R4 075 ✓✓; profit = R1 125 ✓ (3)

1.4.3 Income line through (0 ; 0) and (600 ; 4 800) ✓; cost line through (0 ; 1 800) and (600 ; 3 900) ✓; labelled axes and heading ✓; break-even point (400 ; 3 200) marked ✓ (4)

1.4.4 At 400 vetkoek her income equals her total costs, so she makes neither a profit nor a loss ✓; she must sell more than 400 to make a profit ✓ (2)

1.5 240 000 − 237 100 = 2 900 ✓; tax = 42 678 + 0,26 × 2 900 ✓ = R43 432 ✓; minus rebate: 43 432 − 17 235 ✓ = R26 197 per year ✓; per month = R2 183,08 ✓ (6)

1.6.1 A: R199 ✓; B: 59 + 100 × 1,20 = R179 ✓ (2)

1.6.2 A: R199 ✓; B: 59 + 150 × 1,20 = R239 ✓ (2)

1.6.3 Contract B – it is R20 cheaper for 100 minutes ✓ (accept A if the student argues that usage may increase) (1)

[35]

QUESTION 2

2.1.1 Area = 10 × 8 = 80 m^2 ✓; 25 mm = 0,025 m ✓; volume = 80 × 0,025 = 2 m^3 ✓ = 2 000 ℓ ✓ (4)

2.1.2 V = 3,14 × $0{,}8^2$ × 1,6 ✓✓ = 3,215 m^3 = 3 215,36 ℓ ✓ (3)

2.1.3 Yes – the tank holds about 3 215 ℓ, which is more than 2 000 ℓ (if it is empty) ✓ (1)

2.2.1 100 mm = 0,1 m ✓; V = 4 × 3 × 0,1 ✓ = 1,2 m^3 ✓ (3)

2.2.2 1,2 × 7 = 8,4 → 9 bags ✓✓; 9 × R105 = R945 ✓ (3)

2.3 $\text{Litres} = \frac{640}{100} \times 7{,}5$ ✓✓ $= 48\ \ell$ ✓; $\text{cost} = 48 \times 23{,}45$ ✓ $= \text{R}1\ 125{,}60$ ✓ (5)

2.4.1 15×18 ✓ $= 270\ \text{mg}$ ✓ (2)

2.4.2 $120\ \text{mg in } 5\ \text{m}\ell = 24\ \text{mg per m}\ell$ ✓✓; $270 \div 24$ ✓ $= 11{,}25\ \text{m}\ell$ ✓ (4)

[25]

QUESTION 3

3.1.1 $\text{t} = 145 \div 90$ ✓ $= 1{,}61\ \text{h}$ ✓ $= 1\ \text{hour } 37\ \text{minutes}$ ✓ (3)

3.1.2 $07{:}15 + 1\ \text{h } 37\ \text{min}$ ✓ $= 08{:}52$ ✓ (2)

3.2 $10\ \text{km} = 1\ 000\ 000\ \text{cm}$ ✓; $4 : 1\ 000\ 000$ ✓ $= 1 : 250\ 000$ ✓ (3)

3.3 Any TWO: traffic or roadworks ✓; stops to pick up passengers / rest stops ✓; bad weather; speed limits in towns (2)

[10]

QUESTION 4

4.1.1 $280; 300; 320; 330; 350; 380; 410; 430; 460; 500; 520; 540$ ✓ (1)

4.1.2 $4\ 820 \div 12$ ✓ $= 401{,}67\ \text{kWh}$ ✓ (2)

4.1.3 $\text{Median} = (380 + 410) \div 2 = 395\ \text{kWh}$ ✓✓; $\text{Q1} = (320 + 330) \div 2 = 325\ \text{kWh}$ ✓; $\text{Q3} = (460 + 500) \div 2 = 480\ \text{kWh}$ ✓ (4)

4.1.4 $\text{Range} = 540 - 280 = 260\ \text{kWh}$ ✓; $\text{IQR} = 480 - 325 = 155\ \text{kWh}$ ✓ (2)

4.1.5 Number line with a scale ✓; box from 325 to 480 ✓; median line at 395 ✓; whiskers to 280 and 540 ✓ (4)

4.1.6 May to August (winter months) ✓ – more heating, longer nights and hot water use in winter ✓ (2)

4.2 The differences between the bars look much bigger than they really are ✓, so readers may think sales grew dramatically when the change is small ✓ (2)

4.3 A population is the whole group being studied ✓; a sample is a smaller part chosen from it ✓; e.g. all students at a college (population) and 50 students who answered a questionnaire (sample) ✓ (3)

[20]

QUESTION 5

5.1.1 First branch: faulty $\frac{3}{10}$, working $\frac{7}{10}$ ✓; second branch after faulty: faulty $\frac{2}{9}$, working $\frac{7}{9}$ ✓; after working: faulty $\frac{3}{9}$, working $\frac{6}{9}$ ✓; all outcomes labelled ✓ (4)

5.1.2 $\frac{3}{10} \times \frac{2}{9}$ ✓ $= \frac{6}{90} = \frac{1}{15}$ ✓ (2)

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5.1.3 $1 - \frac{1}{15}$ ✓ $= \frac{14}{15}$ ✓ (2)

5.2 $\frac{18}{60}$ ✓ $= 0{,}3\ (30\%)$ ✓ (2)

[10]

TOTAL: 100

END OF MARKING GUIDELINE